Install MongoDB and pymongo

Follow the steps for your computer, in order. It takes about 20 minutes. You need two things: the MongoDB server (the database, which runs in the background) and pymongo (the Python library). At the end there is a test that tells you if everything works, and a table with the most common problems.

Step 0: Check your computer first

Check	How	You need
Windows version	Press Windows key + R, type winver, press Enter	Windows 10 or 11 (64-bit)
Mac version	Apple menu > About This Mac	macOS 14 (Sonoma) or newer. Older Mac: see the section about older versions
Python version	In a terminal: python --version (Mac: python3 --version)	Python 3.9 or newer
Admin rights (Windows)	Can you install programs yourself?	Yes, or ask IT

- Don't install MongoDB inside WSL (Linux on Windows). It isn't supported. Use the normal Windows installer.

Windows 10 and 11

1. **Download**
   Go to mongodb.com/try/download/community. Choose the newest version, platform Windows x64 and package msi. Click Download. If it doesn't install or start on your computer, see 'If the newest version doesn't work' below.
2. **Start the installer**
   Open the .msi file from your Downloads folder. Click Next and accept the license.
3. **Setup type**
   Choose Complete.
4. **Service Configuration**
   Keep 'Install MongoD as a Service' ticked and 'Run service as Network Service user' selected. Don't change the folders. Click Next.
5. **Compass**
   The installer asks about MongoDB Compass. Untick 'Install MongoDB Compass': we don't use it, and it makes the install slower.
6. **Install**
   Click Install, and Yes when Windows asks for admin permission. If a firewall window pops up, click Allow access.
7. **If you get 'Service MongoDB Server failed to start'**
   Click Ignore and let the installer finish. Then open the Start menu, type Services, find MongoDB Server in the list, right-click it and choose Start.

8. **Install pymongo**
   Open the terminal in VS Code (Terminal > New Terminal) and type:

```
py -m pip install pymongo
```

Mac (macOS 14 or newer)

1. **Developer tools**
   Open the Terminal app and type the command below. If it says they are already installed, that's fine.

```
xcode-select --install
```

2. **Homebrew**
   If you don't have Homebrew: go to brew.sh, copy the install command and run it in the Terminal. When it finishes, it shows 'Next steps' with two commands: run them too, otherwise the brew command isn't found.
3. **Add MongoDB's package list**
   Run these three commands, one at a time:

```
brew tap mongodb/brew
brew trust mongodb/brew
brew update
```

4. **Install MongoDB**
   The number at the end is the version. Install the newest one:

```
brew install mongodb-community@9.0
```

5. **Start MongoDB**
   This starts the server now, and every time the Mac starts:

```
brew services start mongodb-community@9.0
```

6. **Check that it's running**
   The line for mongodb-community should say started:

```
brew services list
```

7. **If the Mac blocks it**
   If you get a message that the developer can't be verified: open System Settings > Privacy & Security, scroll down and click Open Anyway next to the message about mongod. Then start it again.
8. **Install pymongo**
   In the VS Code terminal (if you get 'externally-managed-environment', see Common problems):

```
python3 -m pip install pymongo
```

If the newest version doesn't work

- MongoDB 7.0, 8.0 and 9.0 all work with the course code and with pymongo, so an older version is fine.
- Uninstall the version that didn't work first. Windows: Settings > Apps > MongoDB > Uninstall. Mac: brew uninstall mongodb-community@9.0

1. **Windows**
   On the download page, open the Version dropdown and pick an older version: try 8.0 first, then 7.0. Install it exactly like above. Even older versions are under 'Archived Releases' on the download page: 6.0 is the last version that lists Windows 10 as officially supported.
2. **Mac**
   Change the number to an older version, and use the same number when you start it. Try 8.0 first; if that doesn't work, use 7.0 the same way:

```
brew install mongodb-community@8.0
brew services start mongodb-community@8.0
```

Test that everything works

1. **Make a test file**
   In VS Code, make a file called test_mongo.py with these three lines, and run it:

```
from pymongo import MongoClient
client = MongoClient("mongodb://localhost:27017/", serverSelectionTimeoutMS=3000)
print(client.list_database_names())
```

2. **Success**
   If you see ['admin', 'config', 'local'], everything works. You're done!
3. **Not working?**
   Find your error message in the table below.

Common problems

Error message	What it means	Fix
ServerSelectionTimeoutError ... Connection refused	The MongoDB server isn't running	Windows: Start menu > Services > MongoDB Server > right-click > Start. Mac: brew services start mongodb-community@9.0, then check with brew services list
ModuleNotFoundError: No module named 'pymongo'	pymongo was installed for a different Python than the one VS Code uses	In VS Code: Ctrl+Shift+P (Mac: Cmd+Shift+P) > 'Python: Select Interpreter' and note which Python it is. Then install with that same Python: py -m pip install pymongo (Mac: python3 -m pip install pymongo) in the VS Code terminal
'pip' is not recognized (Windows)	pip isn't on the PATH	Use py -m pip install pymongo instead
'python' is not recognized (Windows)	Python isn't on the PATH	Use py instead of python, for example py test_mongo.py
error: externally-managed-environment (Mac)	The Mac's Python doesn't allow pip installs	In your course folder: python3 -m venv .venv, then source .venv/bin/activate, then pip install pymongo. In VS Code, select the .venv interpreter
zsh: command not found: brew (Mac)	Homebrew's 'Next steps' were skipped	Run: eval "$(/opt/homebrew/bin/brew shellenv)" (Intel Mac: /usr/local/bin/brew), or reinstall Homebrew and run the Next steps
No available formula ... mongodb-community	The MongoDB package list is missing	Run brew tap mongodb/brew and brew trust mongodb/brew again, then the install command
The installer says you need admin rights	Your computer is locked by IT	Ask IT to install MongoDB Community Server for you
The installer finishes, but MongoDB never starts	This version doesn't work on your computer	Uninstall it and install an older version (see the section about older versions)